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EXPERIMENT-9

Aim : To understand Docker Architecture and Container Life Cycle, install Docker andexecute docker commands to manage images and interact with containers

Theory :

Docker is a containerization platform that enables developers to package applications and their dependencies into lightweight, portable units called containers. Unlike traditional virtual machines, containers share the host operating system kernel, making them more efficient in terms of resource utilization and faster to start. Docker follows a client-server architecture, where the Docker client communicates with the Docker daemon to build, run, and manage containers.

The Docker architecture consists of key components such as Docker Client, Docker Daemon, Docker Images, and Docker Containers. The client sends commands (e.g., build, run, pull) to the daemon, which performs operations like creating and managing containers. Docker images are read-only templates used to create containers, while containers are running instances of these images. Docker also uses registries like Docker Hub to store and distribute images.

The container life cycle in Docker includes multiple stages: Create, Start, Run, Pause/Unpause, Stop, and Remove. A container is created from an image, started to execute the application, and can be paused or stopped when not in use. It can also be restarted or removed when no longer needed. By executing Docker commands, users can manage images and interact with containers efficiently, enabling consistent development, testing, and deployment across different environments.

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```
C:\Users\Lenovo>docker --help

Usage:  docker [OPTIONS] COMMAND

A self-sufficient runtime for containers

Common Commands:
  run         Create and run a new container from an image
  exec        Execute a command in a running container
  ps          List containers
  build       Build an image from a Dockerfile
  pull        Download an image from a registry
  push        Upload an image to a registry
  images      List images
  login       Log in to a registry
  logout      Log out from a registry
  search      Search Docker Hub for images
  version     Show the Docker version information
  info        Display system-wide information

Management Commands:
  builder     Manage builds
  buildx*     Docker Buildx (Docker Inc., v0.12.1-desktop.4)
  compose*    Docker Compose (Docker Inc., v2.24.6-desktop.1)
  container   Manage containers
  context     Manage contexts
  debug*      Get a shell into any image or container. (Docker Inc., 0.0.24)
  dev*        Docker Dev Environments (Docker Inc., v0.1.0)
  extension*  Manages Docker extensions (Docker Inc., v0.2.22)
  feedback*   Provide feedback, right in your terminal! (Docker Inc., v1.0.4)
  image       Manage images
  init*       Creates Docker-related starter files for your project (Docker Inc., v1.0.1)
  manifest    Manage Docker image manifests and manifest lists
  network     Manage networks
  plugin      Manage plugins
  sbom*       View the packaged-based Software Bill Of Materials (SBOM) for an image (Anchore Inc., 0.6.0)
  scout*      Docker Scout (Docker Inc., v1.5.0)
  system      Manage Docker
  trust       Manage trust on Docker images
  volume      Manage volumes
```

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```
C:\Users\Lenovo>docker --version
Docker version 25.0.3, build 4debf41

C:\Users\Lenovo>
```

```
C:\Users\Lenovo>docker search ubuntu
NAME                            DESCRIPTION                                     STARS     OFFICIAL
ubuntu                          Ubuntu is a Debian-based Linux operating sys…   17789     [OK]
ubuntu/squid                    Squid is a caching proxy for the Web. Long-t…   124
ubuntu/nginx                    Nginx, a high-performance reverse proxy & we…   139
ubuntu/cortex                   Cortex provides storage for Prometheus. Long…   4
ubuntu/bind9                    BIND 9 is a very flexible, full-featured DNS…   117
ubuntu/kafka                    Apache Kafka, a distributed event streaming …   60
ubuntu/zookeeper                ZooKeeper maintains configuration informatio…   14
ubuntu/apache2                  Apache, a secure & extensible open-source HT…   102
ubuntu/prometheus               Prometheus is a systems and service monitori…   78
ubuntu/mysql                    MySQL open source fast, stable, multi-thread…   72
ubuntu/dotnet-aspnet            Chiselled Ubuntu runtime image for ASP.NET a…   26
ubuntu/jre                      Chiseled Java runtime based on Ubuntu. Long-…   24
ubuntu/postgres                 PostgreSQL is an open source object-relation…   42
ubuntu/python                   A chiselled Ubuntu rock with the Python runt…   31
ubuntu/redis                    Redis, an open source key-value store. Long-…   24
ubuntu/dotnet-runtime           Chiselled Ubuntu runtime image for .NET apps…   24
ubuntu/dotnet-deps              Chiselled Ubuntu for self-contained .NET & A…   16
ubuntu/grafana                  Grafana, a feature rich metrics dashboard & …   12
ubuntu/memcached                Memcached, in-memory keyvalue store for smal…   6
ubuntu/prometheus-alertmanager  Alertmanager handles client alerts from Prom…   10
ubuntu/cassandra                Cassandra, an open source NoSQL distributed …   3
ubuntu/mlflow                   MLFlow: for managing the machine learning li…   6
ubuntu/loki                     Grafana Loki, a log aggregation system like …   2
ubuntu/mimir                    Ubuntu ROCK for Mimir, a horizontally scalab…   0
ubuntu/opentelemetry-collector  This is a rock for the OpenTelemetry Collect…   0
```

```
C:\Users\Lenovo>docker search --filter=stars=1000 ubuntu
NAME      DESCRIPTION                                     STARS     OFFICIAL
ubuntu    Ubuntu is a Debian-based Linux operating sys…   17789     [OK]
```

```
C:\Users\Lenovo>docker search mysql
NAME                 DESCRIPTION                                     STARS     OFFICIAL
mysql                MySQL is a widely used, open-source relation…   16070     [OK]
bitnami/mysql        Bitnami Secure Image for mysql                  147
circleci/mysql       MySQL is a widely used, open-source relation…   32
bitnamicharts/mysql  Bitnami Helm chart for MySQL                    0
cimg/mysql                                                           3
ubuntu/mysql         MySQL open source fast, stable, multi-thread…   72
google/mysql         MySQL server for Google Compute Engine          26
linuxserver/mysql    A Mysql container, brought to you by LinuxSe…   48
elestio/mysql        Mysql, verified and packaged by Elestio         3
alpine/mysql         mysql client                                    7
docksal/mysql        MySQL service images for Docksal - https://d…   0
eclipse/mysql        Mysql 5.7, curl, rsync                          3
ilios/mysql          Mysql configured for running Ilios              1
datajoint/mysql      MySQL image pre-configured to work smoothly …   2
ddev/mysql           ARM64 base images for ddev-dbserver-mysql-8.…   2
mirantis/mysql                                                       0
mysql/mysql-server   Optimized MySQL Server Docker images. Create…   1033
corpusops/mysql      https://github.com/corpusops/docker-images/     0
cleanstart/mysql     Secure by Design, Built for Speed, Hardened …   0
vulhub/mysql                                                         1
cbioportal/mysql     This repository hosts MySQL database images …   1
vitess/mysql         Lightweight image to run MySQL with Vitess      1
mysql/mysql-router   MySQL Router provides transparent routing be…   28
mysql/mysql-cluster  Experimental MySQL Cluster Docker images. Cr…   100
nasqueron/mysql                                                      1
```

```
C:\Users\Lenovo>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
01d7766a2e4a: Pull complete
Digest: sha256:d1e2e92c075e5ca139d51a140fff46f84315c0fdce203eab2807c7e495eff4f9
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest

What's Next?
  1. Sign in to your Docker account → docker login
  2. View a summary of image vulnerabilities and recommendations → docker scout quickview ubuntu

C:\Users\Lenovo>
```

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```
C:\Users\Lenovo>docker images
REPOSITORY                        TAG                                       IMAGE ID       CREATED         SIZE
ubuntu                            latest                                    bbdabce66f1b   13 days ago     78.1MB
ubuntu                            <none>                                    802541663949   6 months ago    78.1MB
ubuntu                            <none>                                    ca2b0f26964c   24 months ago   77.9MB
redis                             <none>                                    170a1e90f843   2 years ago     138MB
docker/dev-environments-default   stable-1                                  7eb85f44c229   3 years ago     610MB
crazymax/linguist                 7.20.0                                    dc5f9ae3be6a   3 years ago     72.2MB
docker/desktop-git-helper         5a4fca126aadcd3f6cc3a011aa991de982ae7000  efe2d67c403b   4 years ago     44.2MB

C:\Users\Lenovo>docker run -it ubuntu
root@715e3a4a7ed8:/# exit
exit

C:\Users\Lenovo>docker run -it ubuntu
root@2ddd2c7718ec:/# admin
bash: admin: command not found
root@2ddd2c7718ec:/# ^C
root@2ddd2c7718ec:/# exit
exit

C:\Users\Lenovo>docker run -d httpd
Unable to find image 'httpd:latest' locally
latest: Pulling from library/httpd
0c8d55a45c0d: Pull complete
fe1db8dd446f: Pull complete
4f4fb700ef54: Pull complete
30973b009bab: Pull complete
85b0d820aa56: Pull complete
996228ca33ab: Pull complete
Digest: sha256:b89c19a390514d6767e8c62f29375d0577190be448f63b24f5f11d6b03f7bf18
Status: Downloaded newer image for httpd:latest
```

```
C:\Users\Lenovo>docker run -d mysql
Unable to find image 'mysql:latest' locally
latest: Pulling from library/mysql
74a8e4bbd9fe: Pull complete
357926362d31: Pull complete
f3702ac323ca: Pull complete
658630c937c6: Pull complete
ab1b8ea72826: Pull complete
61941d436e88: Pull complete
c89a19cdad3b: Pull complete
00af5f73db53: Pull complete
8314ff001dec: Pull complete
4a629f1008ff: Pull complete
Digest: sha256:e5dc14f6e01c3e577e669337d2855c6d1561b30d8ef2c592e63e4e8a9a52650a
Status: Downloaded newer image for mysql:latest
af5bbf473e08d0b9a7264168926d9712c94a6d94497687fed89179d31e770ccd

C:\Users\Lenovo>docker run -d ubuntu
17ae4f0c3240c5b5ba81df3d23be8a1eef290d48684f89b47335567685514b7a

C:\Users\Lenovo>docker ps
CONTAINER ID   IMAGE                                   COMMAND              CREATED              STATUS              PORTS     NAMES
9f10642c64c6   httpd                                   "httpd-foreground"   About a minute ago   Up About a minute   80/tcp    wonderful_babbage
a388ed3e2ae7   docker/dev-environments-default:stable-1   "sleep infinity"     23 months ago        Up 12 minutes               siddhesh-app-1
```

Conclusion : Successfully implemented Docker Architecture and Container Life Cycle, install Docker and execute docker commands to manage images and interact with containers